

Shifted monomial moments

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Abstract

First step of the continuous-amplitude milestone (Astra #16, route C: monomial moments, polynomial density, bounded mass). Dressing the bare normal moment integral with a monomial u^γ shifts the exponents $h \mapsto h + \gamma$. Normalising by the scale $N^{-\lambda}(\log N)^{|J|-1}$ of the *undressed* integral: if γ vanishes on the minimal set J , the normalised moment converges to $\Gamma(\lambda)\beta^{-\lambda}/(|J| - 1)! \cdot \prod_{i \in J} \frac{1}{2k_i} \prod_{i \notin J} \frac{1}{h_i + \gamma_i + 1 - 2k_i\lambda}$ (`shiftedMoment_tendsto_of_face`); otherwise it tends to 0 (`shiftedMoment_tendsto_zero`). These are the monomial moments of the face-supported limiting functional. Zero sorry/axiom.

1 The things formalised

```
theorem tendsto_log_pow_div_pow (a b : ℝ) (hab : a < b) :
  Tendsto (fun N => log N ^ a / log N ^ b) atTop ( 0)
theorem ratioExp_le_shift / ratioExp_shift_eq_of_zero / ratioExp_lt_shift / shift_min
theorem shiftedMoment_tendsto_of_face (d) (h k : Fin (d+1) → ℝ) (hk) (l) (hl) (h)
  (hmin) (hatt) ( : Fin (d+1) → ℝ) (h : i, ratioExp h k i = l → i = 0) :
  Tendsto (fun N => monomialBoxReal (d + 1) (fun i => h i + i) k N
    / (N ^ (-1) * log N ^ (multCount (ratioExp h k) l - 1))) atTop
  ( (monomialMixedConst (fun i => h i + i) k l ))
theorem shiftedMoment_tendsto_zero ... (h : j, ratioExp h k j = l → j = 0) :
  Tendsto (fun N => monomialBoxReal (d + 1) (fun i => h i + i) k N
    / (N ^ (-1) * log N ^ (multCount (ratioExp h k) l - 1))) atTop ( 0)
```

2 Mathematics

Shifting never decreases a ratio, strictly increases it exactly where $\gamma_i > 0$. For a face shift the minimum and its multiplicity are unchanged and the mixed theorem (unit 181) applies to $h + \gamma$. Otherwise the multiplicity of λ strictly drops: if it stays positive, the same exponent appears with a lower log power and $(\log N)^{m'-1}/(\log N)^{m-1} \rightarrow 0$; if it drops to zero, every shifted ratio exceeds λ , the new minimum $\lambda' > \lambda$ is attained, and the quotient of power-log leading terms with $\lambda < \lambda'$ tends to zero (`tendsto_quotient_zero_of_lt`).

3 Paper-facing meaning

For the dressed normal moment $\int u^h \eta(u) e^{-Nu^{2k}}$ these limits are the moments of the leading functional: Taylor monomials supported on the nonminimal coordinates $I \setminus J$ contribute at leading order with the residue factors $1/(h_i + \gamma_i + 1 - 2k_i\lambda)$, while any monomial involving a minimal coordinate is of lower order. This is the mechanism behind Astra #16's qualification of eq. `thm_leading_coeff` for mixed ratios.

4 Lean notes

- `set m' := multCount ... with hm'` does not fold occurrences inside hypotheses created *later*; `rw [← hm'] at hT` after obtaining them.
- `unfold` fails on a goal whose `set` variables hide the constant; `rw [hm]` first, then `unfold`.
- `rw [if_neg H]` takes the negation as an argument; the remaining goal is the numeric inequality, not `H`.
- After `unfold ratioExp` the beta-redex `(fun i => h i + i) i` blocks `rw`; use `simp` only.